

RUMETH HERATH

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience building production-grade, cloud-ready software systems across backend, AI, and mobile platforms. Currently working at Advent Atum delivering end-to-end solutions involving Python services, system optimisation, and scalable architectures. Strong foundation in software design, APIs, CI/CD practices, and cross-functional collaboration. Recognised with a Rising Star Award for delivering reliable, real-world systems. Passionate about building robust, scalable solutions that help clients move forward.

EDUCATION

Bachelor of Engineering – Software Engineering

Macquarie University — Sydney, NSW — Feb 2021 – Nov 2025

Relevant Coursework: Software Design, Data Structures, Algorithms, Database Systems, Mobile Application Development

WORK EXPERIENCE

Software Engineer

Advent Atum — Sydney, NSW — Jan 2025 – Present

- Designed and contributed to end-to-end backend and AI pipelines, supporting production deployment of computer vision systems.
- Supported migration of core services to Rust, improving system performance, memory safety, and reliability.
- Collaborated with cross-functional engineering teams to deliver scalable software solutions aligned with business and technical requirements.
- Applied best practices in testing, debugging, and software lifecycle management to improve system stability.

AI Data Engineer (Casual)

Advent Atum — Sydney, NSW — May 2025 – Dec 2025

- Collaborated with engineering team to build Flutter-based Android application integrating AI features, improving usability and performance metrics
- Worked alongside senior engineers to train, test, and deploy machine learning models for real-world drone detection projects
- Enhanced debugging and testing workflows, improving code reliability and reducing production failure rates by implementing systematic testing procedures

AI Software Engineer (Intern)

Advent Atum — Sydney, NSW — July 2024 – March 2025

- Developed and optimized image recognition models and LSTM neural networks in Python using PyTorch framework
- Contributed to new feature development in Flutter Android application, enhancing application functionality and stability
- Collaborated with cross-functional teams using Agile methodologies and version control systems for timely project delivery
- Implemented code reviews and best practices to maintain high code quality standards

Techbar IT Support

Woolworths Group — Sydney, NSW — May 2018

- Provided technical support including operating system re-imaging and device setup for employees
- Ensured smooth system rollouts and maintained hardware/software configurations

TECHNICAL SKILLS

Languages	Frameworks/Tools	Software Engineering	Cloud + DevOps
Java	Flutter	OOP	CI/CD
Python	React, React Native	Debugging	Linux Environments
C, C, C++	PyTorch	Testing	Git workflows
Javascript	Restful APIs	CI/CD	
Rust	MySQL	Agile Development	
HTML, CSS			

PROJECTS

Thesis: Drone Detection for Privacy Preservation

- Designed a modular, end-to-end detection pipeline using YOLOv8 and YOLOv11 for real-time drone and subcomponent detection.
- Built and evaluated computer vision models in Python, focusing on robustness, scalability, and real-world deployment constraints.
- Conducting research on AI-powered surveillance systems and privacy protection technologies

Bioscope Wedge – Image Acquisition System

- Developed software pipeline components for high-resolution microscope image capture and processing
- Integrated with Rust-based MicroCore API and Webschool platform to enable AI-powered digital microscope experience
- Applied system-level thinking to balance accuracy, performance, and privacy considerations in safety-critical applications

Charity Mobile Application

- Built donation and event-tracking application using React Native for cross-platform mobile development
- Implemented user authentication, payment integration, and real-time event management features
- Designed intuitive user interface to enable seamless user engagement and donation processing

Personal Portfolio Website (In Progress)

- Designing responsive personal portfolio website using React and TailwindCSS
- Showcasing software engineering projects, technical skills, and professional experience
- Implementing modern web development best practices and responsive design principles

ACHIEVEMENTS & AWARDS

Rising Star Award – Advent Atum (2025)

Recognized for exceptional contributions to AI projects and machine learning initiatives

Certificate of Best Poster Presentation for Software Engineering (2025)

Macquarie University Final Year Thesis Projects Showcase

Letter of Commendation – COMP1050 HD (2021)

Macquarie University – Session 2

REFERENCES

Available upon request